

Research Planning Report: AI-Based Road Network Extraction from Satellite Imagery

Date: February 26, 2026

Subject: Initial Research Strategy and Work Breakdown Structure

Prepared For: Project Research Team (3 Members)

1. Executive Summary

This report outlines the foundational research strategy for the project on "AI-Based Road Network Extraction from Satellite Imagery." To maximize efficiency and ensure a comprehensive understanding of the current state-of-the-art literature, the research phase has been divided into three distinct, non-overlapping domains. This division ensures the team covers spatial identification (pixels), network connectivity (topology), and experimental validation (data and metrics) without redundancy.

2. Research Allocation & Focus Areas

The literature review is distributed among the three team members as follows:

Domain 1: Pixel-Level Segmentation Architectures (Assigned to: Member 1)

Objective: Investigate the core deep learning models responsible for classifying and extracting road pixels (Semantic Segmentation) from high-resolution imagery.

Key Research Topics:

- **Foundational Architectures:** Evolution and application of CNNs, specifically U-Net and its variants (e.g., ResUNet).
- **State-of-the-Art Models:** In-depth review of **D-LinkNet** (DeepGlobe challenge winner) and its structural advantages for linear feature extraction.
- **Vision Transformers (ViTs):** The integration of attention mechanisms (e.g., SegFormer, Swin Transformers) to capture global context and long continuous road structures.
- **Multi-Scale Feature Extraction:** Methodologies for simultaneously identifying wide highways and narrow residential streets.

Target Keywords: "Semantic segmentation satellite imagery roads", "U-Net variants road extraction", "Transformer based road network extraction", "D-LinkNet".

Domain 2: Topology-Aware and Graph-Based Models (Assigned to: Member 2)

Objective: Research advanced techniques designed to preserve the connectivity, geometry, and topological accuracy of road networks, moving beyond raw pixel masks to functional graphs.

Key Research Topics:

- **Graph Neural Networks (GNNs):** Utilizing nodes (intersections) and edges (road segments) for network prediction.
- **Iterative Tracing Algorithms:** Analysis of models like **RoadTracer**, which formulate road extraction as a sequential decision-making process.
- **Direct-to-Graph/Vector Models:** Reviewing architectures like **Sat2Graph** that output vectorized road networks directly.
- **Topology-Preserving Constraints:** The use of specialized loss functions (e.g., connectivity loss) to penalize fragmented road predictions during model training.

Target Keywords: "Topology-aware road extraction" , "Graph Neural Networks road network" , "Sat2Graph" , "Iterative road tracing satellite" , "Vectorized road extraction" .

Domain 3: Datasets, Domain Adaptation, and Evaluation (Assigned to: Member 3)

Objective: Become the subject matter expert on ground-truth data, the complexities of real-world satellite imagery, and the rigorous metrics required to validate the model's performance.

Key Research Topics:

- **Standard Datasets:** Comparative analysis of major datasets including DeepGlobe, SpaceNet (specifically v3 and v5), and the Massachusetts Roads Dataset.
- **Advanced Evaluation Metrics:** Researching specialized routing metrics such as **APLS (Average Path Length Similarity)** and **TOPO metrics**, which evaluate functional connectivity rather than just pixel overlap (IoU/Jaccard Index).
- **Occlusion & Environmental Challenges:** Methodologies for handling shadows, cloud cover, tall buildings, and dense tree canopies.
- **Domain Adaptation/Transfer Learning:** Techniques for maintaining model accuracy when deployed on geographically diverse imagery (e.g., models trained on US data tested on Asian topographies).

Target Keywords: "SpaceNet road extraction dataset" , "APLS metric road network" , "Domain adaptation satellite imagery" , "Occlusion handling road extraction" .

3. Execution Strategy and Timeline

The research phase will be executed over an estimated 5-to-6-week timeline, following a structured workflow to integrate individual findings into a cohesive project plan.

Phase 1: Foundation (Week 1)

- **Action:** All members read 2-3 high-level survey/review papers on deep learning for remote sensing.

- **Deliverable:** Shared understanding of baseline terminology (Multispectral imaging, GIS, IoU, basic CNNs).

Phase 2: Divergent Deep Dive (Weeks 2-4)

- **Action:** Members independently research their assigned domains, focusing on papers published from 2020 onward.
- **Deliverable:** A collaborative Literature Review Matrix (spreadsheet) tracking: *Paper Title, Year, Core Innovation, Dataset Utilized, Model Strengths, and Limitations*. Target: 5-8 foundational papers per member.

Phase 3: Knowledge Sync (Weekly, Concurrent with Phase 2)

- **Action:** A mandatory 1-hour weekly sync where each member presents a 15-minute briefing on their most significant finding of the week.
- **Deliverable:** Cross-functional understanding of how architectures (Domain 1) impact topology (Domain 2) and are measured by specific metrics (Domain 3).

Phase 4: Convergence & Project Definition (Week 5)

- **Action:** Review the synthesized literature matrix as a team.
- **Deliverable:** A finalized Project Proposal detailing the chosen architecture, the dataset to be utilized, the specific loss functions to be implemented, and the evaluation metrics that will define the project's success.